

Reviewer Pack — Birkhoff Spectral Gap of Gate Co-occurrence Bounds AC^0 PARITY...

Entry 14ed7020b118 · INCONCLUSIVE

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Birkhoff Spectral Gap of Gate Co-occurrence Bounds AC^0 PARITY Correlation

Entry ID: 14ed7020b118 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-18 05:16:35 UTC

1. Conjecture

Field A (mathematical branch): Birkhoff polytope / doubly-stochastic spectral theory (Sinkhorn 1964; Linial-Samorodnitsky-Wigderson 2000; Forbes-Saxena 2018 on permanent approximation) applied as a STRUCTURAL invariant of the gate-pair input co-occurrence matrix of a Boolean circuit. The $\text{Sym}(n)$ -equivariant Sinkhorn scaling and its λ_2 spectral gap are a non-ring convex-analytic invariant (the scaling map uses divisions and row/column normalisations that do not extend to $\mathbb{F}_q$, so it cannot algebrize). An arXiv search for ‘Sinkhorn scaling’ AND (‘ AC^0 ’ OR ‘switching lemma’ OR ‘circuit complexity’) returns 0 direct hits and <5 adjacent papers, all on permanent approximation in algebraic complexity, never on AC^0 PARITY correlation. **Field B** (complexity object): $AC^0[d]$ circuit correlation with PARITY_n in the Håstad 1986 switching-lemma / Razborov-Smolensky regime, in the Cook-Reckhow-Krajíček bounded-arithmetic framework where AC^0 PARITY size lower bounds correspond to $V^0 \not\sqsubseteq V^0(\oplus)$ separations. The invariant is a STRUCTURAL property of the circuit DAG (not of its truth table), shielding it from the Razborov-Rudich natural-proofs barrier; it is $\text{Sym}(n)$ -equivariant under the natural action permuting input variables, fulfilling the focus’s ‘natural with respect to a non-trivial group action’ requirement.

Statement:

For an AC^0 circuit C on n inputs of depth d and size s , let $I(g) \subseteq [n]$ be the input cone of gate g (variables reaching g in C). Define $A(C) \in \mathbb{R}^{\{n \times n\} \times \{ \geq 0 \}}$ by $A[i,j] = \#\{ \text{gates } g : \{x_i, x_j\} \subseteq I(g) \}$, set $\tilde{A} = A + 10^{-3} \cdot J$, and $\kappa(C)$ satisfies $\kappa(C) \leq \exp(-c \cdot n \cdot g(C) / d)$. A single AC^0 circuit C with $\kappa(C) > \exp(-c \cdot n \cdot g(C) / d)$ (for $c = 1/8$) refutes the conjecture.

Rationale (proposer’s reasoning):

Hastad’s switching lemma kills AC^0 correlation with PARITY via random restrictions that spread mass evenly across input coordinates; the Sinkhorn-doubly-stochastic projection captures the SAME spreading geometrically — $g(C)$ measures how ‘mixed’ the gate-pair co-occurrence matrix is under symmetric scaling, and large $g(C)$ forces no input pair to dominate the circuit’s structure. Because $D(C)$ transforms equivariantly under $\text{Sym}(n)$ permuting inputs, $g(C)$ is a representation-theoretic invariant of the circuit’s wiring, not of its truth table — so it cannot be a Razborov-Rudich natural property. The construction is purely convex-analytic (Sinkhorn uses divisions, not ring operations) and so does not algebrize under polynomial extensions of the Boolean ring, sidestepping the AVOID examples (tropical homotopy stability, tropical weight accumulation) that hit ALGEBRIZATION.

Taxonomy category: BOUNDED_ARITHMETIC (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 07afc7557bfe3c27

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

Across ≥ 2000 AC⁰ trials ($n \in \{8, 12, 16\}$, $d \in \{2, 3\}$, $s \in \{6, 12, 24, 48\}$, 30 seeds/cell), conjecture is SUPPORTED iff every trial satisfies $\kappa(C) \leq \exp(-n \cdot g(C)/(8d))$ AND Spearman $\rho(\kappa, \exp(-n \cdot g/(8d))) \geq 0.3$; FALSIFIED iff any trial yields $\kappa(C) > 1.05 \cdot \exp(-n \cdot g(C)/(8d))$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.95	SAFE	SAFE
NATURAL_PROOFS	SAFE	0.95	SAFE	SAFE
ALGEBRIZATION	SAFE	0.95	SAFE	SAFE
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Sinkhorn scaling AC⁰ circuit lower bounds PARITY correlation - doubly stochastic spectral gap Boolean circuit complexity natural proofs - Birkhoff polytope gate co-occurrence matrix switching lemma AC⁰

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=5.5s

5.1 Generated Python source

```
import sys
import random
import math
import itertools
from collections import defaultdict

def matrix_multiply(A, B):
    n = len(A)
    m = len(B[0])
    p = len(B)
    result = [[0.0 for _ in range(m)] for _ in range(n)]
    for i in range(n):
        for j in range(m):
            for k in range(p):
                result[i][j] += A[i][k] * B[k][j]
    return result
```

```

def matrix_transpose(A):
    return [list(row) for row in zip(*A)]

def matrix_add(A, B):
    return [[A[i][j] + B[i][j] for j in range(len(A[0]))] for i in range(len(A))]

def matrix_scale(A, scalar):
    return [[A[i][j] * scalar for j in range(len(A[0]))] for i in range(len(A))]

def matrix_normalize_rows(A):
    row_sums = [sum(row) for row in A]
    return [[A[i][j] / row_sums[i] if row_sums[i] != 0 else 0.0 for j in range(len(A[0]))] for i in range(len(A))]

def matrix_normalize_cols(A):
    A_T = matrix_transpose(A)
    A_T_normalized = matrix_normalize_rows(A_T)
    return matrix_transpose(A_T_normalized)

def sinkhorn(A, iterations=25):
    for _ in range(iterations):
        A = matrix_normalize_rows(A)
        A = matrix_normalize_cols(A)
    return A

def power_iteration(A, num_iterations=100):
    n = len(A)
    b = [1.0 / n for _ in range(n)]
    for _ in range(num_iterations):
        b_new = [sum(A[i][j] * b[j] for j in range(n)) for i in range(n)]
        norm = math.sqrt(sum(x**2 for x in b_new))
        b = [x / norm for x in b_new]
    return b

def compute_eigenvalue(A, b):
    Ab = [sum(A[i][j] * b[j] for j in range(len(b))) for i in range(len(A))]
    return sum(Ab[i] * b[i] for i in range(len(b)))

def compute_second_eigenvalue(A):
    n = len(A)
    b1 = power_iteration(A)
    lambda1 = compute_eigenvalue(A, b1)
    A_shifted = [[A[i][j] - lambda1 * (i == j) for j in range(n)] for i in range(n)]
    b2 = power_iteration(A_shifted)
    lambda2 = compute_eigenvalue(A_shifted, b2)
    return lambda2

def build_co_occurrence_matrix(C, n):
    A = [[0 for _ in range(n)] for _ in range(n)]
    for gate in C:
        inputs = gate['inputs']
        for i in range(len(inputs)):
            for j in range(i + 1, len(inputs)):
                x_i = inputs[i]
                x_j = inputs[j]
                A[x_i][x_j] += 1

```

```

        A[x_j][x_i] += 1
    return A

def compute_g(C, n):
    A = build_co_occurrence_matrix(C, n)
    J = [[1.0 for _ in range(n)] for _ in range(n)]
    A_tilde = matrix_add(A, matrix_scale(J, 1e-3))
    D = sinkhorn(A_tilde)
    lambda2 = compute_second_eigenvalue(D)
    return 1.0 - abs(lambda2)

def evaluate_circuit(C, x):
    values = list(x)
    for gate in C:
        inputs = gate['inputs']
        if gate['type'] == 'AND':
            values.append(all(values[i] for i in inputs))
        elif gate['type'] == 'OR':
            values.append(any(values[i] for i in inputs))
    return values[-1]

def compute_parity_correlation(C, n):
    total = 0.0
    for x in itertools.product([0, 1], repeat=n):
        parity = sum(x) % 2
        C_x = evaluate_circuit(C, x)
        total += (-1) ** (C_x ^ parity)
    return abs(total) / (2 ** n)

def generate_random_circuit(n, d, s):
    C = []
    for _ in range(s):
        layer = random.randint(0, d - 1)
        gate_type = random.choice(['AND', 'OR'])
        num_inputs = random.randint(1, min(n, 2))
        inputs = random.sample(range(n), num_inputs)
        C.append({'type': gate_type, 'inputs': inputs})
    return C

def run_trial(seed):
    random.seed(seed)
    n = random.choice([8, 12, 16])
    d = random.choice([2, 3])
    s = random.choice([6, 12, 24, 48])
    C = generate_random_circuit(n, d, s)
    g = compute_g(C, n)
    kappa = compute_parity_correlation(C, n)
    bound = math.exp(-n * g / (8 * d))
    conjecture_holds = kappa <= bound
    counterexample = "" if conjecture_holds else f"n={n}, d={d}, s={s}, kappa={kappa}, bound={bound}"
    return {
        "metric_name": "kappa_bound_ratio",
        "metric_value": kappa / bound if bound > 0 else float('inf'),
        "instances_tested": 1,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample,
    }

```

```

        "n": n,
        "d": d,
        "s": s,
        "g": g,
        "kappa": kappa,
        "bound": bound
    }

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97]
    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    metric_values = [r["metric_value"] for r in results]
    mean_metric = sum(metric_values) / len(metric_values) if metric_values else 0.0
    std_metric = math.sqrt(sum((x - mean_metric) ** 2 for x in metric_values) / len(metric_values))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results) if results else 0.0

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric} std={std_metric} support_fraction={support_fraction}")
    else:
        failing_trials = [r for r in results if not r["conjecture_holds"]]
        if failing_trials:
            first_failing_seed = seeds[results.index(failing_trials[0])]
            print(f"RESULT: FALSIFIED counterexample={failing_trials[0]['counterexample']}")
        else:
            print("RESULT: INCONCLUSIVE reason=empty_results")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

jecture_holds': True, 'counterexample': '', 'n': 16, 'd': 3, 's': 48, 'g': -
0.7359093566514678, 'kappa': 0.0, 'bound': 1.6333060917151618}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.8870418116493577, 'kappa': 0.0, 'bound': 1.8064456904906498}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.8660207193645115, 'kappa': 0.0, 'bound': 1.5418921941310963}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.805622159349614, 'kappa': 0.0, 'bound': 1.4960242355796014}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.9261501129555745, 'kappa': 0.0, 'bound': 1.5889525991259912}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.8006633967206949, 'kappa': 0.0, 'bound': 1.4923196155245912}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.7322580901888673, 'kappa': 0.0, 'bound': 2.07977162120796}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': '

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test is vacuous: every reported kappa is exactly 0.0, so any non-negative bound trivially holds — this is textbook metric saturation / trivial sub-case. Worse, many reported $g(C)$ values are negative (e.g., -0.96, -0.89), which is impossible for a true doubly-stochastic matrix where $|\lambda_2| \leq 1$; this signals a construction-gap bug (Sinkhorn didn't actually converge in 25 iterations, or λ_2 is being computed on the wrong matrix), making the 'bound' meaningless. $n \leq 16$ with random tiny circuits (

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Every trial reports $\kappa=0.0$ exactly, so the inequality holds vacuously; additionally $g(C)$ is frequently negative, which is impossible for a genuine doubly-stochastic matrix and indicates a Sinkhorn/eigenvalue bug. The pre-registered criterion (≥ 2000 trials, Spearman $\rho \geq 0.3$) is not demonstrated, and the regime tested ($n \leq 16$, tiny random circuits) never approximates PARITY. | next: Fix the Sinkhorn scaling (verify convergence and that $|\lambda_2(D)| \leq 1$), then test on circuits designed to have nontrivial PARI

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	claude_max	opus	0	0	296717
2	preregistration	claude_max	opus	0	0	5660
3	novelty	claude_max	opus	0	0	3368
4	novelty	claude_max	opus	0	0	7223
5	test_gen	mistral	codestral-latest	0	0	316152
6	test_gen	mistral	codestral-latest	0	0	314521
7	test_gen	mistral	codestral-latest	0	0	313479
8	test_gen	mistral	codestral-latest	0	0	313045
9	critic	claude_max	opus	0	0	16637
10	judge	claude_max	opus	0	0	6306

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 1593109 ms total latency. Provider mix: {'claude_max': 6, 'mistral': 4}

(full prompt+response transcripts available in `research/audit/14ed7020b118.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/14ed7020b118.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/14ed7020b118.tar.gz (if generated)